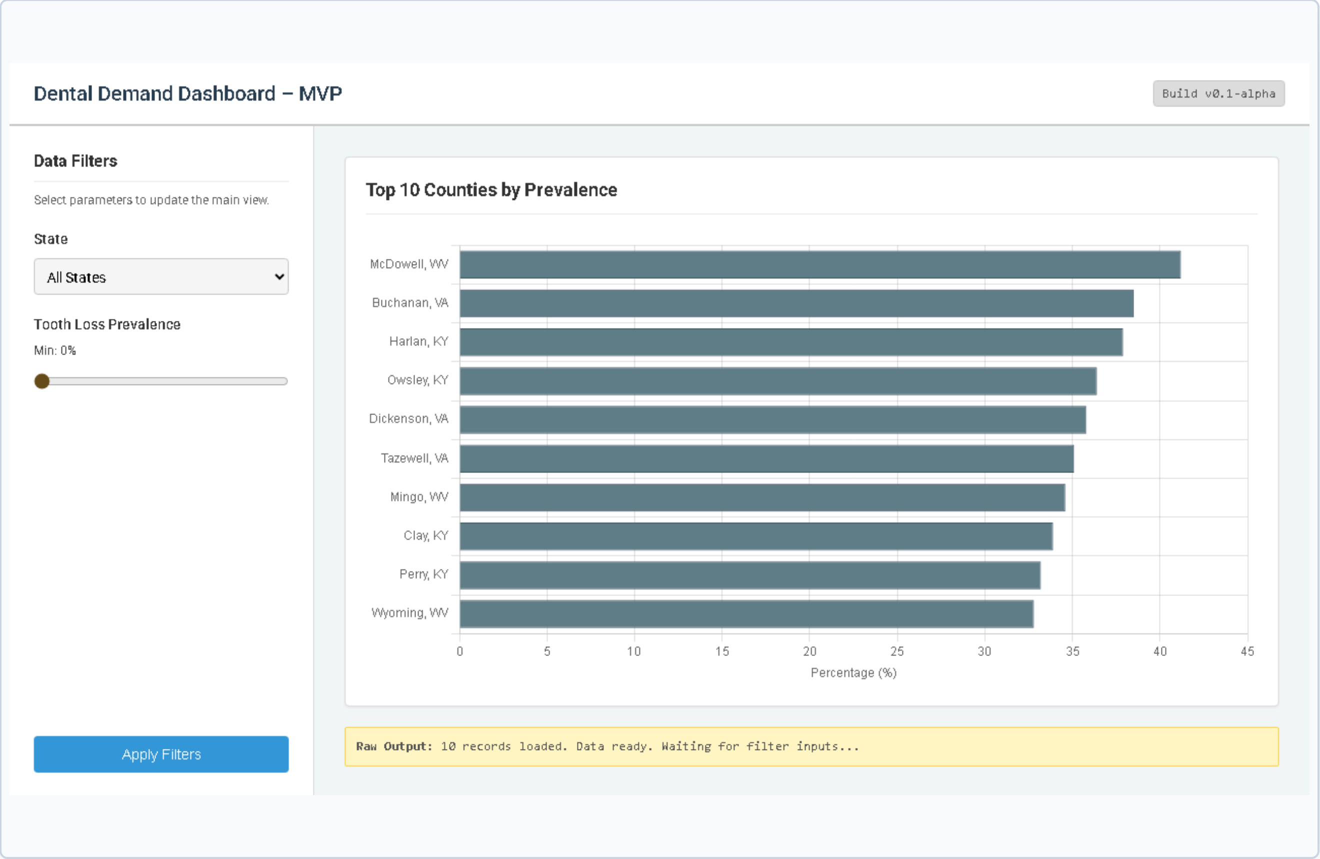
Team: Amirtheshwaran Navaneethakrishnan, Hruthik Narayan Sarva, Anderson Zhu

MVP Demo Streamlit Prototype

Baseline Application

Designed for rapid deployment and testing in secondary markets.

- Features interactive state selection and prevalence thresholding using range sliders.
- Visualizes the Top 10 counties by absolute complete tooth loss volume using actual CDC metrics.
- Successfully identifies high-need regions for early-stage expansion planning.



Visual Grammar Justification

1. Chart Selection

Horizontal Bar Chart

The optimal selection for ranking categorical data (Counties). It accommodates long text labels without requiring awkward rotation.

2. Visual Marks

Quantitative Magnitude

Solid geometries anchored at zero provide an unambiguous physical representation of the underlying value.

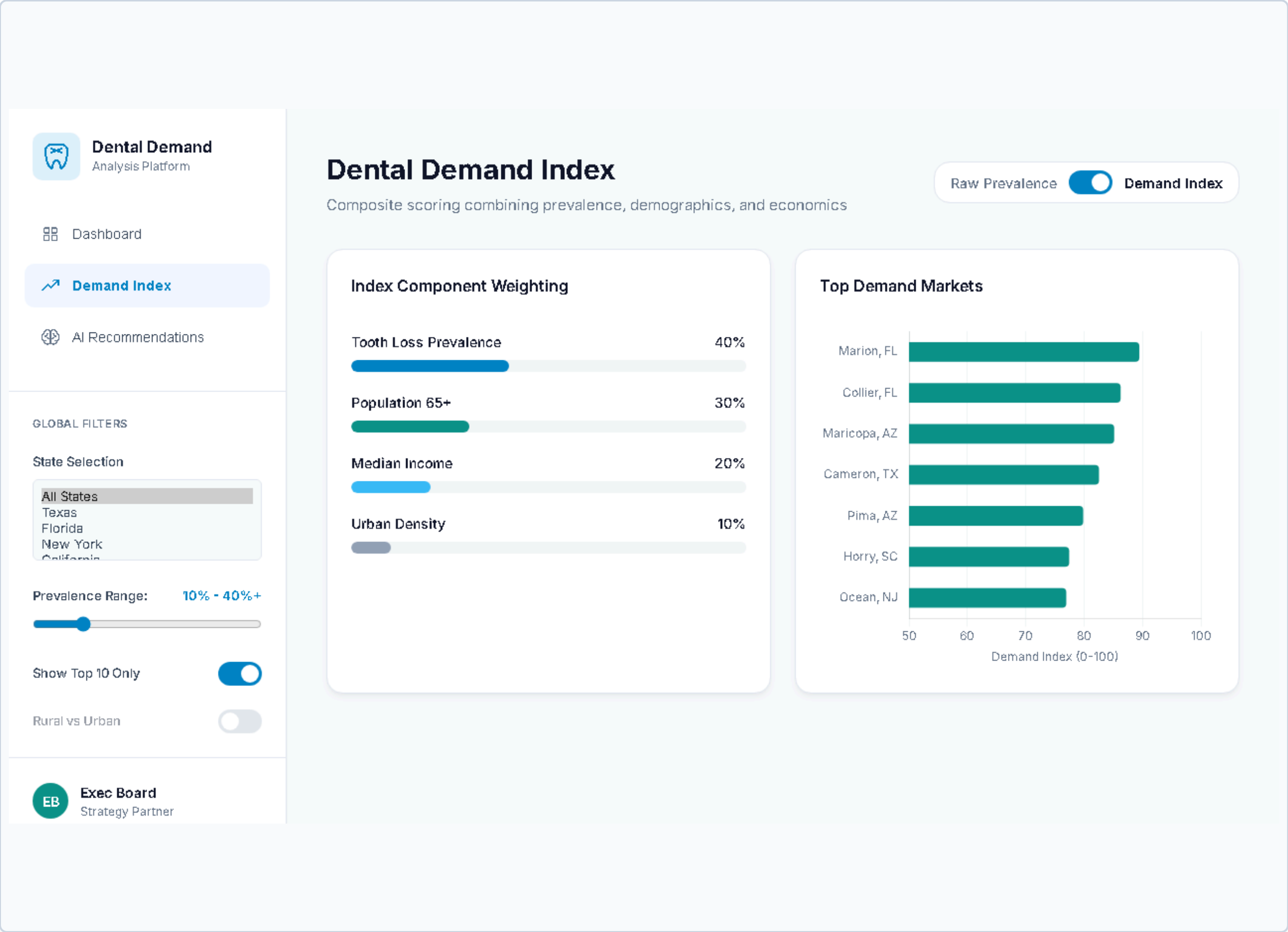
3. Data Channels

- **Length:** Encodes prevalence percentage. Length is the most accurate visual channel for human perception.
- **Position:** Enables easy sequential and comparative reading from top to bottom.

4. Effectiveness

- Provides immediate identification of highest-need areas.
- Imposes near-zero cognitive load on the executive user.
- Gridlines are minimized to maximize the data-ink ratio.

Phase 2 Expansion Intelligence



Enhanced Capabilities

Implemented in Streamlit with weighted scoring toggle. Integration of advanced metrics moving beyond raw prevalence.

- Combines tooth loss, population 65+, income demographics, and urbanization into a single weighted score.
- **User-adjustable weights via sliders** for live recomputation of ranking.
- Transitions the platform from a descriptive visualization to an analytical scoring system.

AI Integration Strategy

Data Inputs → Feature Engineering → Clustering Model → Prediction Engine → Recommendations

- **Pattern Recognition:** The system clusters geographic regions based on hidden demographic and geospatial patterns.
- **Typology Prediction:** Predicts the optimal clinic type (such as a Denture Center vs. an Implant Studio) for market fit.
- **Actionable Outputs:** Delivers prescriptive location and strategy recommendations mapped directly to anticipated ROI.

Sprint to Final Implementation

Remaining Work

- Implement operational AI model (clustering + prediction).
- Connect the backend model directly to the front-end UI interface.
- Enable real-time inference and on-the-fly threshold adjustments.

UI Improvements

- Improve application responsiveness across desktop viewports.
- Add micro-interactive feedback to buttons and charts.
- Enhance overall visual polish and finalize branded components.

Final Goal

Deliver a fully integrated decision-support system that accelerates executive strategy formation and minimizes expansion risk.